

6.0 EPILEPSY

Epilepsy is a chronic non-communicable disease of the brain that affects around 50 million people worldwide. It is characterized by recurrent seizures, which are brief episodes of involuntary movement that may involve a part of the body (partial) or the entire body (generalized) and are sometimes accompanied by loss of consciousness and control of bowel or bladder function.

Seizure episodes are a result of excessive electrical discharges in a group of brain cells. Different parts of the brain can be the site of such discharges. Seizures can vary from the briefest lapses of attention or muscle jerks to severe and prolonged convulsions. Seizures can also vary in frequency, from less than one per year to several per day.

6.1 Symptoms of epilepsy

Seizures can affect people in different ways, depending on which part of the brain is involved.

Possible symptoms include:

- Uncontrollable jerking and shaking, called a "fit".
- Losing awareness and staring blankly into space.
- Becoming stiff.
- Strange sensations, such as a "rising" feeling in the tummy, unusual smells or tastes, and a tingling feeling in your arms or legs.
- Collapsing.

6.2 Different Types of Epilepsy

Healthcare providers classify epilepsies by their seizure type. Seizure categories are based on where they start in your brain, your level of awareness during a seizure and by presence or absence of muscle movements.

There are two major seizure groups:

6.2.1 Generalized onset seizures

General onset seizures affect a widespread network of cells on both sides of your brain at the same time. There are six types of generalized seizures.

- **Absence seizures:** This seizure type causes a blank stare or “staring into space” (a brief loss of awareness). There may be minor muscle movements, including eye blinking, lip-smacking or chewing motions, hand motions or rubbing fingers. Absence seizures are more common in children, last for only seconds (usually less than 10 seconds) and are commonly mistaken for daydreaming. This seizure type used to be called petit mal seizures.
- **Atonic seizures:** Atonic means “without tone.” An atonic seizure means you’ve lost muscle control or your muscles are weak during your seizure. Parts of your body may droop or drop such as your eyelids or head, or you may fall to the ground during this short seizure (usually less than 15 seconds). This seizure type is sometimes called “drop seizure” or “drop attack.”
- **Tonic seizures:** Tonic means “with tone.” A tonic seizure means your muscle tone has greatly increased. Your arms, legs, back or whole body may be tense or stiff, causing you to fall. You may be aware or have a small change in awareness during this short seizure (usually less than 20 seconds).
- **Clonic seizures:** “Clonus” means fast, repeating stiffening and relaxing of a muscle (“jerking”). A clonic seizure happens when muscles continuously jerk for seconds to a minute or muscles stiffen followed by jerking for seconds up to two minutes.
- **Tonic-clonic seizures:** This seizure type is a combination of muscle stiffness (tonic) and repeated, rhythmic muscle jerking (clonic). Healthcare providers may call this seizure a convulsion, and once called it a grand mal seizure. Tonic-clonic seizures are what most people think of when they hear the word “seizure.” You lose consciousness, fall to the ground, your muscles stiffen and jerk for one to five minutes. You may bite your tongue, drool and lose muscle control of bowels or bladder, making you poop or pee.

- **Myoclonic seizures:** This seizure type causes brief, shock-like muscle jerks or twitches (“myo” means muscle, “clonus” means muscle jerking). Myoclonic seizures usually last only a couple of seconds.

6.2.2 Focal onset seizures

Focal onset seizures start in one area, or network of cells, on one side of your brain. This seizure used to be called partial onset seizure. There are two types of focal seizures:

- **Focal onset aware seizure** means you’re awake and aware during the seizure. Healthcare providers once called this a simple partial seizure. Symptoms may include:
 - Changes in your senses — how things taste, smell or sound.
 - Changes in your emotions.
 - Uncontrolled muscle jerking, usually in arms or legs.
 - Seeing flashing lights, feeling dizzy, having a tingling sensation.
- **Focal onset impaired awareness seizure** means you’re confused or have lost awareness or consciousness during the seizure. This seizure type used to be called complex partial seizure. Symptoms may include:
 - Blank stare or a “staring into space.”
 - Repetitive movements like eye blinking, lip-smacking or chewing motion, hand rubbing or finger motions.

6.3 Taking Care of Children with Epilepsy

Managing children with epilepsy involves a combination of medication, lifestyle modifications, and sometimes surgery or other interventions. Most children with epilepsy live full, active lives that include school, friends, sports, and other activities. How much epilepsy interrupts a child's life depends upon the kind of epilepsy, the success of treatment, and many other factors.

- **Get an accurate diagnosis:** Diagnosing epilepsy in children poses different challenges than with adults. Depending on the type and severity of a child's epilepsy, the symptoms of certain seizure types may appear as harmless daydreaming – the kind of outward absentmindedness we think of as common in children. The child's care team will conduct a number of medical exams, including an EEG and possibly MRI exams. To provide more information and help develop a more effective treatment plan, you may also be asked to take detailed notes or even a video of the child during a seizure. The goal is to achieve a seizure-free state when possible to minimize medication side-effects, and to enhance quality of life for the child and family.
- **Monitor the child's medicine.** “Not all children are placed on anti-epileptic medication, a decision that is made in consultation with the child's pediatrician and neurologist,” Dr. Constantinou says. If the child is placed on epilepsy medication, some questions to ask include:
 - How often should they take the medication?
 - Should they take it at the same time each day?
 - What should I do if they forget to take it?
 - Are there any potential side effects?

Make sure the child's school or daycare knows about the epilepsy medication, and that arrangements are made to take it at school.

- **Avoid known seizure triggers:** Monitor the child’s daily activities and write down any seizure triggers. Every child with epilepsy is different, but some common triggers include:
 - Stress
 - Missed medication
 - Lack of sleep
 - Infrequently flashing lights

It is important to know that most children with epilepsy have no issue watching TV, viewing computer screens or playing video games.

- **Recognize the warning signs of a seizure:** There are many kinds of seizures, all with varying degrees of severity and different warning signs. These warning signs may include:
 - Excessive staring.
 - Jerking or twitching movements.
 - Periods of rapid eye blinking.
 - Stiffening of the body and extremities.
 - Shallow or accelerated breathing, or breathing stops.
 - Loss of bowel or bladder control.
 - Not responding to noise or words.
 - Appearing hazy or confused.
 - Loss of consciousness.
 - Excessive nodding or head swaying.
- **Emphasize the positive.** “Our children look to us for cues on how to act, and this is even more true for a child with epilepsy, who may struggle to cope,” says Dr. Constantinou. Stay positive and avoid discussing any burdens – financial or otherwise – the child’s epilepsy places on you or the family. A positive attitude can have a tremendous effect on your child’s mood and overall outlook about his or her epilepsy.
- **Watch for changes in behavior or mood:** For some children, having epilepsy and taking anti-epileptic drugs may affect their behavior or mood. If you notice any of the following, consult with the child’s medical team or parents:

- Learning problems or trouble in school.
 - Irritability.
 - Withdrawing from friends and family.
 - Eating less.
 - Negative statements.
 - Trouble sleeping.
- **Demystify epilepsy:** Encouraging the child to talk about epilepsy may help them feel better. In addition, discuss seizures and epilepsy openly with the child and family. Make sure everyone understands as much about seizures as possible and is able to assist in an emergency.
- **Encourage normal activities:** Let the child be a child. Encourage the child to have as much social contact with other children as possible, and to participate in the activities approved by their pediatrician and neurologist. Focus on the things the child can do, and encourage them to be active participants in life. Having epilepsy may create some restrictions, but in general, the child should be able to take part in most activities.
- **Recognize the child's epilepsy could change over time:** In many cases, children outgrow their epilepsy as they enter their teens. If the child is taking epilepsy medication and has been seizure-free for a number of years, the child's care team may suggest slowly stopping the medication.

6.4 What to do when a child is having a seizure.

Learn about Seizure First Aid



Watching a child have a seizure can be upsetting and frightening. Even more concerning for most parents is the chance the child could have a seizure when you're not there. The first step is to learn what to do if the child has a seizure. The next is to teach others who may be with the child when you're not.

The most important thing to do when a child has a seizure is keep the child safe. Often, you don't need to do anything but wait for the seizure to end. A child might just have a short staring spell and then go on as if nothing happened. Or a child could suddenly fall, but then just need a little comforting or help getting up.

6.4.1 What to Do If a Child Has a Generalized Seizure

If a child has a "grand mal" or tonic-clonic seizure (cries, falls, stiffens and then starts moving in jerking, uncontrolled movements), here are the things to remember:

DO:

- Move anything that could hurt the child.
- Put something soft and flat under the child's head (pillow, folded jacket or sweater).
- Turn the child gently to one side to prevent choking.
- Loosen anything around the child's neck.

Keep track of how long the seizure lasts and other important information to tell the doctor. Stay calm and comfort the child afterward.

DON'T:

- Try to hold the child.
- Try to stop any movements or tongue-biting.
- Try to open or put something in the child's mouth.

6.4.2 Continuous Seizures

Prolonged or cluster seizures — seizures that don't stop after five minutes or continue one after another — can turn into status epilepticus, which means a “continuous state of seizure.” This requires immediate medical attention. It is good to have written instructions from the doctor to give to the paramedics or emergency room staff in case the child ever experiences a continuous seizure. You should also know where the closest and best hospital is.

If the child has a continuous seizure:

- Call an ambulance. Do not try to move the child unless there is no other choice.
- Ask the doctor to call the emergency room and let them know you're coming.
- If you have written instructions from the doctor for the doctors and paramedics, make sure you have them.